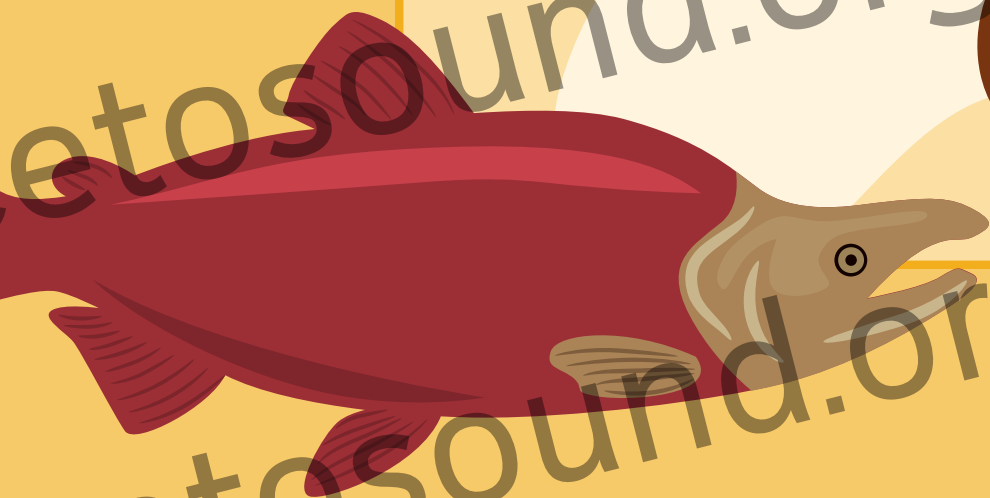


BUILD A SALMON REDD

Can you build a streambed that
keeps salmon eggs safe?



SALMON LIFE CYCLE

Adult Salmon



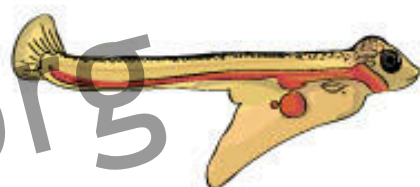
Fry/Juvenile



Egg



Alevin



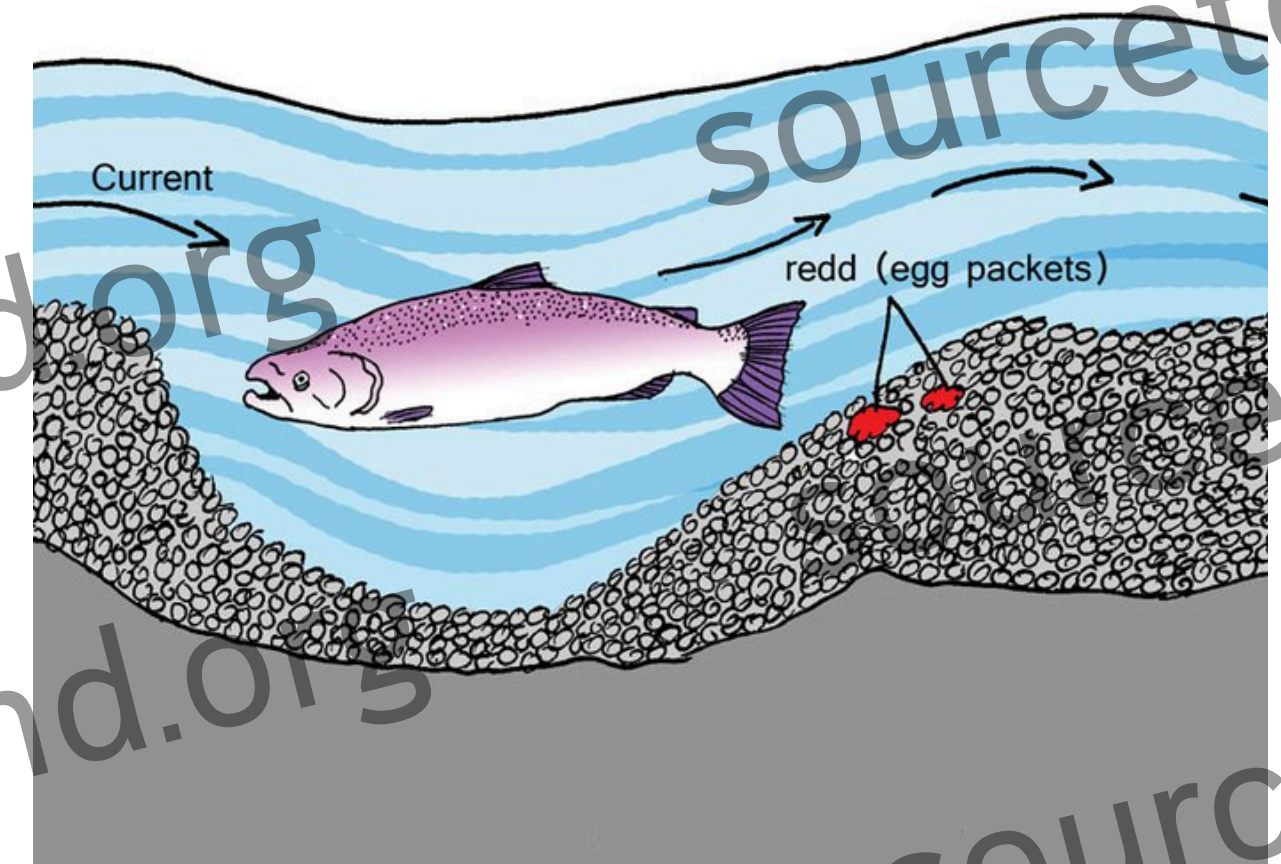
Where do
you think
salmon leave
their eggs?

WHAT IS A REDD?

Redd: a nest that female salmon make in stream gravel to hold and protect her eggs

HOW?

1. Female chooses a spot with clean, oxygen-rich gravel
2. Rapid tail beats to create a depression or "pocket" in the gravel
3. Deposit eggs into the water
4. dig trenches around to focus water flow on the redd



BUILD THE BEST REDD!



OBJECTIVES

1. Keep all 5 eggs inside
 2. Keep the eggs from washing away
 3. Leave space for water to move through
- You do NOT need to use every material



DIRECTIONS

1. Choose one end to pour water
2. Watch and observe what happens to the streambed and eggs
3. Record your observations and the amount of eggs that
 - a. stayed protected
 - b. moved

TEST #1: STREAM FLOW



TEST #2: SEDIMENT STORM

DIRECTIONS

1. Add the same amount of fine soil upstream
2. Pour the same amount of water again
3. Watch and observe what happens as sediment travels

DISCUSSION: WHAT CHANGED?

STREAM FLOW

1. How many eggs stay protected?
2. Where did water move?
3. What happened to your streambed?

SEDIMENT STORM

1. Where did sediment collect?
2. Did it fill spaces between rocks?
If so, how?
3. What happened to the eggs? Did they move?

IF YOU COULD REBUILD...

What would you change?



1. Which materials would you use more/less of?
2. How would you change the layout of gravel?
3. Where would you position the eggs for best protection?

FINAL TAKEAWAY

What happens upstream matters downstream

Erosion upstream means sediments can enter streams and fill gravel spaces, reducing water flow around eggs and making salmon habitats less suitable



How can we help protect salmon?

- Plant trees and native plants near streams
- Keep dirt and sediment out of storm drains
- Reduce polluted runoff

